

Paper title

Author names

¹ affiliation 1

² affiliation 2

³ affiliation 3

* Corresponding author: xxx

Keywords: Tuberculosis, vaccine, clinical trial, modelling, sample size, TASA, IGRA.

Abstract

xxxxx

Introduction

Introductory paragraph. How can we make TB vaccine trials more efficient?

Background on TB (including fast/slow progression) and TB vaccine candidates/ current vaccine trials

Paragraph on the research problem - vaccine trials are slow/inefficient/large, need to target fast progressors, new test for recent infection, how might this new test be useful

Paragraph outlining what this article does ("Using XXX, this article estimates ..."). How many people do we need to enrol for an effective TB vaccine trial, under different circumstances?

Material and methods

Model details

Statistical model of vaccine trial *recruitment*. Two approaches - analytic and stochastic. In both cases, our objective function is to achieve an 'effective' trial, defined as continually recruiting until at least 50 fast progressors have been enrolled. This value is motivated by power calculations [ref]. Individuals are sampled from the population aged 18-50 years, with a South Africa-like population age distribution (UN WPP 2024 estimates) following an ongoing clinical trial [ref UN WPP and m72]. Requirements for enrollment in the model are as follows: Asymptomatic (vaccine is given prior to disease); AND Infected (to increase the chance of progressing to disease during a ~3y trial). Note there is no vaccination in the model; we simulate trial *recruitment* only.

The analytic approach uses fundamental probability theory to estimate e.g. $P(\text{asyp} \& \text{infected} \& \text{fast} \mid \text{age } a)$, $P(\text{infected} \& \text{fast} \mid \text{asyp} \& \text{age } a)$, and $P(\text{fast} \mid \text{infected} \& \text{asyp} \& \text{age } a)$. From these, we can calculate

the expected number needed to sample, test, and enrol for an effective trial (50 fast progressors). See Supplementary for a full derivation. The stochastic approach on the other hand simulates random draws from probability distributions, following the below algorithm:

```
while (n_fast < 50) {
  Draw a person of random eligible age from the population.
  Draw their progression status (slow/ fast).
  Draw their time to infection tinf.
  Draw their time to symptoms tsymp.
  If tsymp < age, person is asymptomatic so we test them. Add 1 to n_overall.
  If tinf < age, person is +ve so recruit them. Add 1 to n_recruit.
  If fast, add 1 to n_fast. }
```

tinf is calculated using a survival function with hazard rate rho equal to the force of infection blah. In the unifirom force of infection case, this is approximated as blah. tsymp is then calculated as tinf + blah.

Explain sigma bit (TASA/IGRA) and figure model-flowchart.

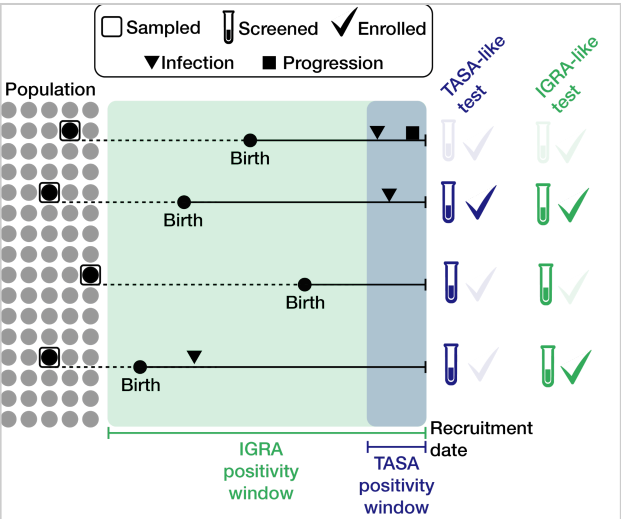


FIGURE-model-flowchart:

Model parameters

Parameter values used in our model are shown in Table model-parameters. The analytic approach uses a uniform force of infection and uniform probability of being a fast progressor. The stochastic approach additionally explores an age-varying force of infection and age-varying probability of being a fast progressor. Rate of progression to disease as a fast (slow) progressor once infected, mu_fast (mu_slow), is assumed to be constant.

TABLE-model-parameters:

Parameter	Description	Base value	Range explored	Reference
minage	Minimum age for trial eligibility	18 years	-	[ref m72]
maxage	Maximum age for trial eligibility	49 years	-	[ref m72]
rho	Force of infection	4%		Fitted parameter

p_fast	Probability of being a fast progressor	5%		
p_slow	Probability of being a slow progressor	1 - p_fast	1 - p_fast	
mu_fast	Rate of progression to disease, for fast progressors	Exponential(1.5)	-	
mu_slow	Rate of progression to disease, for slow progressors	0.0001 per year		
sigma	Duration of positivity of test	80 years	0.5-80	
trial_length	Duration of simulated trial	3 years	-	[ref m72]

Model fitting and validation procedures

Age-varying force of infection, rho, was chosen to emulate ... plausible literature estimates of infection prevalence by age, annual risk of Tuberculosis infection, and annual case data [refs].

Discuss the meshing of analytic and stochastic - the two approaches align with uniform age structure. Other stuff to discuss here?

Results

How would the size of vaccine trial change for each length of test from what we have now (ever infected) all the way down to ~6 months (very recently infected)? Describe result and main takeaways (figure main-plot).

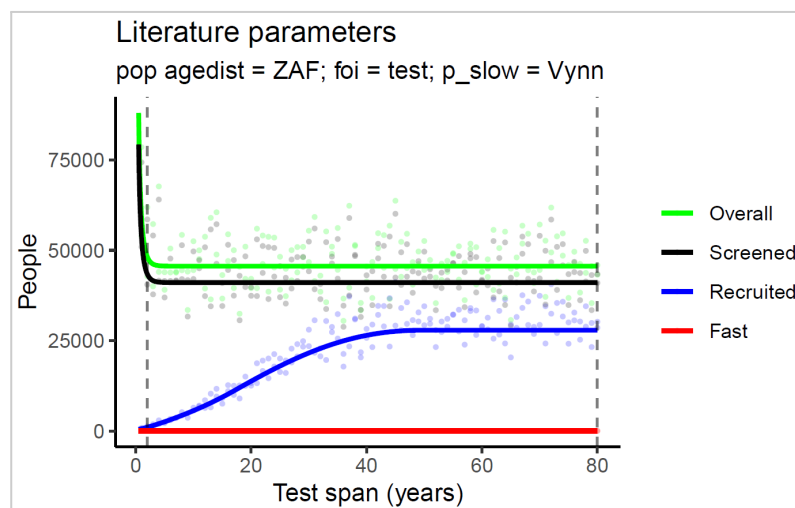


FIGURE main-plot:

Describe quick economic result / cost-saving possibilities. Cost saving estimates for each model scenario are shown in Supplementary Table economic-costs.

TABLE economic-costs

Sensitivity analysis

Additional sensitivity analysis scenarios are shown in the Supplementary.

Discussion

Focusing on this class of new tech that can identify recent infection. These open up various exciting avenues, including the possibility of speeding up vaccine trials - a longstanding issue. Discuss main findings and interpretation.

Paragraph/s to compare to other literature

Miscellaneous discussion points - list here as we think of them

Limitations paragraph

Concluding remarks

Acknowledgements

SMK is supported by a Mathematical Modeling of Policy Options for Evolving Public Health Challenges award from the U.S. National Science Foundation and Centers for Disease Control and Prevention (DMS-2436340). xxx

Conflicts of interest: SMK has received consulting fees from Moderna Therapeutics. xxx. All other authors declare no conflicts of interest.

References